Change Request CR-001: Assign a technician in the manual path

Illustrative change request for the Customer Outage Request case study. It documents a known simplification in the executable process and analyzes the options to fix it.

Field	Value
ID	CR-001
Title	Assign a technician in the manual path
Raised by	Oleksiy Milner (business analyst)
Date	2026-10-23
Type	Defect in process design (found during execution)
Priority	Should (the demo runs without it, but the result is inconsistent)
Status	Proposed

1. Problem

When the skills check fails (HTTP 422), the process routes to the manual task **Clarify and manually Assign to Technician**. That task does not assign anyone. The process then continues through **Start Activity** and **Complete Activity**, so the activity ends `completed` while still assigned to the bucket.

Evidence from the executable demo (database after both runs):

Activity	Path	Status	Assigned to
6	Happy path	completed	technician 31
7	Manual path	completed	resource 25 (bucket)

This contradicts the intent of US-5 (escalated work gets manually assigned) and US-6 (assignment follows the skill rule), and leaves completed work attached to a queue instead of a technician.

2. Proposed outcome

After a mismatch or escalation, the dispatcher enters a technician. The system retries the reassignment with the same skill validation (RULE-3). Only a successful assignment lets the process continue to Start and Complete.

3. Options

Option	Description	Assessment
A. Loop back (recommended)	The manual task collects a `technicianId` and the flow returns to **Reassign Activity To Technician**. A repeated mismatch returns to the dispatcher.	Reuses the existing reassignment call and its validation; no API change; consistent end state.
B. Separate assignment call	After the manual task, add another service task that calls `PATCH` with the dispatcher's choice.	Duplicates the reassignment step and needs its own error handling.
C. End the process after the manual task	The process ends with "Assigned manually"; handling continues outside it.	Avoids the wrong end state, but status tracking stops and the activity stays in the bucket.
D. Leave as is	Keep the documented simplification.	No effort; inconsistent data remains.

Recommendation: Option A.

4. Impact analysis (Option A)

Area	Impact	Change	Effort (estimate)
Requirements	New requirement and updated acceptance criteria	Add FR-14: after a mismatch or escalation the dispatcher can enter a technician and the system retries the reassignment (Must). Extend US-5 with that criterion. Partially answers OQ-2.	Small
BPMN process	Flow and task changes	Link a form with a numeric field `technicianId` to the manual task. Add a flow from the manual task back to Reassign. Remove the "Technician is Assigned" gateway, as the happy path no longer merges.	Medium
OpenAPI spec	None	`PATCH` already accepts `resourceId` and documents 422.	None
Data model / ERD	None	No new entities or attributes.	None
Mock service	None	Skill validation already runs on every reassignment.	None
Tests	New scenarios	Mismatch, then manual entry of a qualified technician, ends with resource 31 and status `completed`. A second mismatch returns to the manual task.	Small
Documentation	Updates	Remove the known simplification from the executable README, add new evidence, and update the traceability table.	Small

Estimated total effort: about half a day (assumption).

5. Risks and dependencies

Risk	Mitigation
Repeated mismatches create a loop	Each loop needs human input; add a dispatcher option to cancel the request if needed.
A new process version does not change running instances	Instances started before deployment finish on the old version; run new tests on new instances.
The form must be linked correctly to the user task	Verify by completing the task in Tasklist and checking the `technicianId` variable.

No external dependencies.

6. Acceptance criteria for the change

- Given `technicianId` 32 at start, when the dispatcher enters 31 in the manual task, then the activity is assigned to technician 31 and finishes `completed`.
- Given the dispatcher enters an unqualified technician, then the response is 422 and the process returns to the manual task.
- After any run, no completed activity remains assigned to a bucket.

7. Traceability

Affects US-5, US-6, FR-5, FR-6, FR-10, RULE-3 and OQ-2, and the BPMN steps **Reassign Activity To Technician** and **Clarify and manually Assign to Technician**.

8. Approval

Role	Name	Decision	Date
Requestor	Oleksiy Milner	Submitted	2026-10-23
Reviewer	Development team lead	Pending	
Approver	Service manager	Pending	